B.TECH DBMS PROJECT

Hostel Accommodation and Student Services Management System

A practical database system for connected, reliable hostel operations



THE PROBLEM

Manual hostel administration creates avoidable risk

MANUAL

Scattered records

- Paper records make it difficult to track students, rooms, visitors, complaints, and maintenance in one place.
- Common failures include duplicate bed allocations, incorrect fee records, overcapacity, and delayed complaint handling.

COMPUTERIZED

One digital system

The website prototype provides a single digital system for faster, more reliable hostel operations.

SYSTEM SCOPE

One system connects every hostel service

Each module supports a complete view of accommodation and student activity.



Student records

Registration and profile management.



Accommodation

Blocks, rooms, and beds.



Allocation history

Allocations, transfers, and vacating records.



Fees

Fee schedules and payment tracking.



Service activity

Visitors, complaints, and maintenance requests.



Reports

Vacancy and occupancy visibility.

DATA MODEL

The ER model mirrors the hostel's real-world structure

Hostel Block → Room One block contains many rooms.	**Room → Bed** One room contains many beds.	**Student → Allocation → Bed** Students receive allocations.
Allocation → Transfer; Allocation → Vacating Record Allocations can lead to transfers or vacating records.	**Fee Schedule → Payment** Operational entities connect student activity with financial records.	**Visitor · Complaint · Maintenance Request** Operational entities connect student activity with visitor, complaint, and maintenance records.

RELATIONAL DESIGN

The relational schema protects consistency

Keys and constraints turn the ER model into dependable records that can be updated without losing relationships.

Integrity first

Core relationships

Table	Connects through
Room	Block ID
Bed	Room ID
Allocation	Student ID + Bed ID
Payment	Fee Schedule ID

Rules that prevent errors

- Primary keys identify each record.
- Foreign keys prevent invalid references.
- Constraints prevent duplicate active allocations and unmatched payments.
- Normalization reduces repeating data.

END-TO-END WORKFLOW

From registration to vacating, each workflow leaves a record

The system preserves a traceable history at every operational checkpoint.

01

Register

Capture required student profile details.

02

Allocate

Check availability before assigning a bed.

03

Record events

Track fees, payments, visitors, complaints, and maintenance.

04

Update status

Reflect transfers and vacating in occupancy records.

05

Report

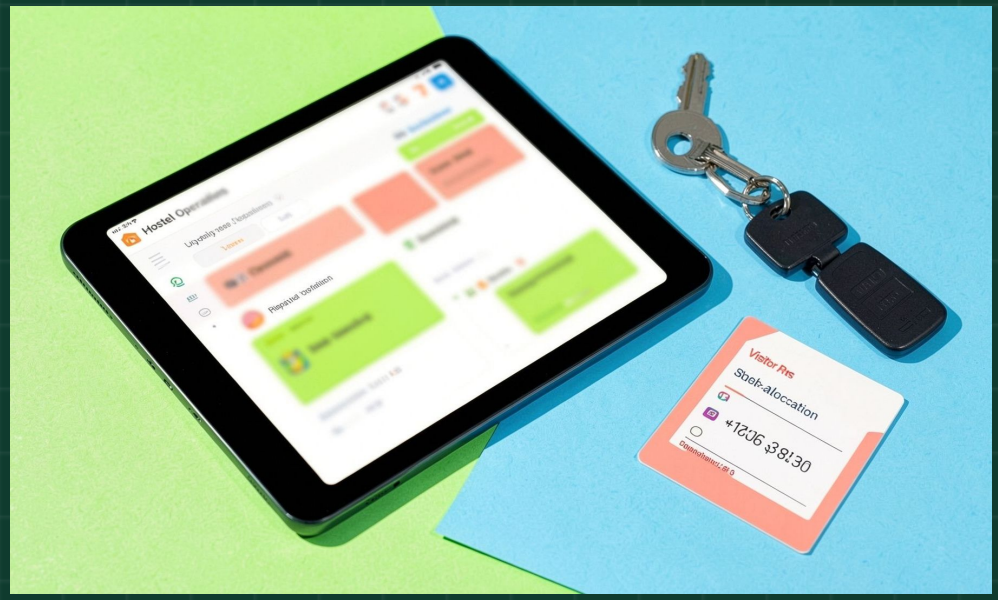
Generate vacancy, occupancy, and service-status reports.

USER INTERFACE

The website turns database operations into simple user actions

- Dashboard: occupancy, vacancies, pending payments, complaints, and maintenance status.
- Admin modules: students, blocks, rooms, beds, allocations, fees, visitors, and service requests.
- Forms validate inputs before records are saved to the database.
- Tables and filters help staff find records quickly and update operational status.

Use editable interface mockups or prototype screenshots to demonstrate the implemented website. If screenshots are added later, replace the mockups without changing the slide structure.



DASHBOARD

Occupancy, vacancies, pending payments, complaints, and maintenance status.

ADMIN MODULES

Students, blocks, rooms, beds, allocations, fees, visitors, and service requests.

PROJECT TAKEAWAY

A reliable database improves hostel service delivery.

Centralized records reduce duplication and improve visibility across hostel operations. Integrity rules support accurate allocation, fee, occupancy, and service histories. The prototype demonstrates how DBMS concepts become a practical student-services application.

Future scope: role-based access, notifications, online payments, analytics, and mobile support.

Student Name | Roll Number | Department | College Name | Guide Name

